Specific heat

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use mass, material and temperature change to calculate heating energy.

SUCCESS CRITERIA - I can...

1. Identify ΔT rather than final temperature alone.
2. Use Q=mcΔT with matching units.
3. Compare materials through specific heat.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “specific heat” and “heat” to explain your model or result. Sentence starters are optional.

specific heat: energy per mass needed for a one-degree temperature change

heat: energy transferred because of a temperature difference

temperature: a measure of thermal state; it is not an energy amount

01 Warm-up: bring back an earlier idea

A 3 kg cart descends 5 m from rest.
Use g=10; losses are 6 J.
Find bottom speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $Q = mc\,\Delta T$; $\Delta T = T_f - T_i$.
No phase change. A temperature difference of 1°C equals 1 K.

Worked reasoning: A 2 kg sample with c=100 J/(kg·°C) warms by 3°C. Heat absorbed is Q=mc ΔT=600 J, with no phase change or work transfer in this model.

Your turn: Draw an energy arrow into the sample and show which three factors produce 600 J.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial T / phase: _____ Final T / phase: _____



Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two equal masses receive the same energy. Material A has a larger specific heat than B. Which has the smaller temperature rise?

Sentence starter: Both … . They differ in … because … .

A: larger c

B: smaller c

Each outline represents the labeled sample or container.

Marzano check - circle: 1 2 3

04 Apply the physics

A heater transfers 2700 J into a 0.20 kg metal sample initially at 18°C. Its specific heat is 450 J/(kg·°C). Assume no phase change or energy loss.
Label the model and find the temperature rise and final temperature.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Why is multiplying by the final temperature instead of ΔT a mistake?

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart starts from rest at height 4 m. Can it reach 6 m with no external energy input or losses? Justify using kinetic energy.
Show the energy changes and transfers; label units.

Word help: kinetic energy = energy associated with motion

Sentence starter: Under the assumption …, the evidence … supports … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 0.50 kg sample needs 4200 J to warm 4°C.
Find c and state a necessary assumption.
Label the energy arrow and identify what changes in the sample.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.5 kg material absorbs 900 J and warms 6°C.
Find c. How much heat warms 1.2 kg of it by 15°C?
Label the energy arrow and identify what changes in the sample.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3